

Quadrotor UAV Kinematics and Dynamics: A Comprehensive Model with Suggested Control Techniques

Abdul Rahman Abdul Majid*
Dept. of Electrical Engineering,
College of Engineering, University
of Sharjah.
Sharjah, UAE.
U19105906@sharjah.ac.ae

Abstract—This paper offers a comprehensive and systematic approach to developing the mathematical model for quadrotor unmanned aerial vehicles (UAVs), focusing on both kinematics and dynamics. Using the well-established Newton-Euler method, the paper simplifies the modeling process to provide a technically sound yet accessible framework that can be applied across various flight modes. Unlike many previous studies that concentrate on specific flight conditions or control systems, this work focuses purely on presenting a complete guide to deriving the quadrotor model, making it a valuable reference for researchers and engineers looking to understand quadrotor behavior in a detailed and structured manner. Although this paper does not propose or implement new control systems, it suggests a few relevant control strategies that could be used with the provided model, considering the complex dynamics of quadrotors. These strategies serve as a foundation for future research in control system development, highlighting the model's utility as a key tool in advancing UAV technology. This work fills a gap in existing literature by offering a thorough and practical guide to quadrotor modeling, which is essential for both theoretical study and real-world applications.

Keywords—Unmanned Aerial Vehicle (UAV) Modeling, Kinematics and Dynamics of Quadrotor drone, Nonlinear Dynamics, Modeling and Control Quadrotor Robot, Newton-Euler Method for Modeling, Control Strategy for Quadrotor System.

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have become increasingly integral to both civilian and military operations, thanks to advancements in sensor technology and communication systems. These UAVs are categorized by size, mass, flight duration, altitude, wing type (fixed-wing or rotor-based), and operating platform. Among them, the quadrotor stands out due to its mechanical simplicity, compact size, ability to hover, maneuverability, and capability for vertical take-off and landing. These attributes make quadrotors ideal for research and various real-time applications, including surveillance, mapping, aerial cinematography, search and rescue missions, agriculture, transportation, communication, and security [1]. They are also particularly effective in hazardous environments such as disaster areas (COVID-19 containment zones in 2020 and others), chemical spill sites, gas leaks, and nuclear power stations. Their versatility extends to precision agriculture and surveillance activities, underscoring their significant market potential. Fig. 1 illustrates the estimated revenue from the quadrotor market based on a report by Price Waterhouse Coopers (PwC) [1].

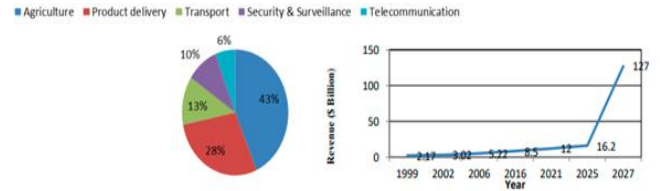


Fig. 1 Estimated revenue for Quadrotor UAVs [1]

In recent years, the control systems for quadrotor UAVs have advanced significantly. Multiple approaches have been developed to manage the position and attitude of quadrotors, ranging from intelligent methods to linear and nonlinear control strategies [2]. A decade-long review from 2009 to 2019 identified thirty-three control approaches for multi-rotor vehicles, including quadrotors [2]. Another review specifically examines quadrotor attitude control strategies [3]. However, controlling quadrotor systems is challenging due to their under-actuated, nonlinear, and highly coupled nature. Additionally, their lightweight design makes them vulnerable to external disturbances, especially during outdoor flights. Traditional PID controllers, although simple and widely used in industrial settings, are insufficient for flying vehicles due to dynamic changes and disturbances like wind. Advanced control strategies employed to address these challenges include sliding mode control (SMC) for time-varying systems [4], [5], robust adaptive control for environments with unknown disturbances [6], the second-order sliding-mode super-twisting algorithm (STA) [7], nonlinear controllers for path-tracking [8], optimal controllers using the LQ regulator method [9], Integral Backstepping (IB) controllers [10], and Neural Network (NN) controllers [11], Active Disturbance Rejection Control (ADRC) [12], [13].

Developing a model is the essential first step for control and simulation. Typically, system models are derived from first principles, physical insights, and sometimes from input-output data. The latter two approaches fall under system identification. For quadrotors, first principles, inherited from traditional flight mechanics, provide a clear dynamic model that elucidates the relationship between inputs and outputs. However, this approach often results in a complex and highly nonlinear mathematical formulation, posing challenges for controller design. As an alternative, system identification effectively derives models by constructing mathematical representations of dynamic systems from observed input-output data—a field that has evolved since 1965, offering numerous methods. Despite these advancements, many

challenges and open problems remain [14], [15], [16], [17]. Historically, most modeling and identification efforts have focused on fixed-wing and helicopter UAVs rather than quadrotors or multirotor [14], [17], likely due to the comparatively complex dynamics of quadrotors. These dynamics exhibit distinctive features in modeling and identification schemes, as various studies have highlighted [14], [15], [16], [17]. Consequently, researchers continue to propose new techniques for system identification and modeling of quadrotor systems in journal papers, conferences, and dissertations [18] [19], [20], [21], [22], [23], [24], [25], [26].

The full nonlinear equations governing quadrotor dynamics have been derived using classical Euler-Newtonian mechanics and Lagrange equations, incorporating various forces, particularly aerodynamic forces, as shown in previously mentioned references. However, a comprehensive and systematic model specifically for quadrotors remains elusive. Quadrotor dynamics display unique characteristics across different flight modes, including VTOL, horizontal flight, aggressive maneuvers, and hovering. Currently, the hovering mode receives the most attention due to its critical importance. Nonetheless, more research should be directed toward other modes, such as aggressive maneuvers, to fully understand and exploit quadrotor capabilities. This paper is motivated to present a comprehensive and systematic kinematics and dynamics model suitable for all flight modes. This paper aims to:

- Provide a comprehensive mathematical model for the quadrotor system, covering both kinematics and dynamics.
- Offer a complete guide and tutorial for deriving the quadrotor's mathematical model.
- Present a systematic representation of the quadrotor system using the Newton-Euler method, which is technically well-established and simpler compared to other approaches.
- Propose optimal and suitable control strategies based on the developed model.

This paper is organized as follows: Section 2 provides a system description, Section 3 presents the mathematical modeling covering kinematics, dynamics, and control of the quadrotor system, Section 4 represents control strategies for quadrotor system, and the final section concludes the study.

II. SYSTEM DESCRIPTION

To understand the motion of the quadrotor robot, it is essential to have the concept of six-degree of freedom (6-DOF) in terms of the equation. From that, it is possible to get the orientation and the position of the quadrotor robot in three dimensions (3-D). Where the change in the orientation and the position over time is the dynamic of the quadrotor robot. The dominant methods for modeling the dynamics of robot systems are the Hamilton method, Euler-Lagrange method, and Newton-Euler method. The Newton-Euler formalism is preferable due to its ease to be understood and approved physically. So, this paper depends on the Newton-Euler formalism for describing the dynamics of the system.

It is possible to describe the position of a robot in terms of three coordinates space, but it is a challenging case for controlling the quadrotor robot without knowing the

orientation of the robot in space. So, it is required to know six coordinates of the vehicle body in space to describe the position. From this the concept of 6-Degree of Freedom originated and emerged, Fig. 2 shows the 6-DOF system. It could be describing the generalized 6-DOF coordinates for the quadrotor robot as:

$$q(x, y, z, \psi, \theta, \phi) \in R^6 \quad (1)$$

Where this generalized form consists of the position of the mass center of the quadrotor relative to Earth-Frame (inertial frame) and three Euler angles. (x, y, z) represents the position of the mass center. It is possible to define it as a translation coordinate.

$$\xi = q(x, y, z) \in R^3 \quad (2)$$

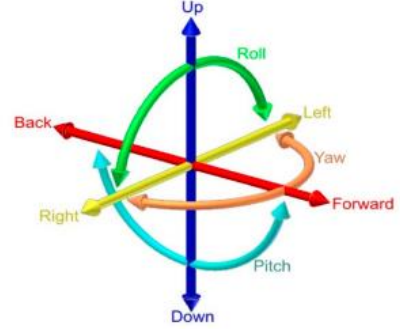


Fig. 2. Six-Degree Freedom

Where (ψ, θ, ϕ) represents Euler-angles. It is possible to define it as a rotational coordinate.

$$\eta = q(\psi, \theta, \phi) \in R^3 \quad (3)$$

Where ψ, θ, ϕ are yaw, pitch, and roll angles respectively. The Newton-Euler method must define two frames of reference. The first one is the Body-fixed frame (B-frame) which is define the right-hand coordinate for body, quadrotor, $(B = x_B, y_B, z_B)$ and the correspondent body linear velocity vector $\vec{v}_B = [u \ v \ w]^T$ and its angular rate vector $w_B = [p \ q \ r]^T$, as shown in Fig. 3 It is also denoted as a reference frame, and it is shown in Fig. 3 as a North-East-Down reference frame (O_{NED} -reference frame). The second one is Earth-fixed inertial frame (E-frame) also it's called the navigation coordinate system ($E = x_E, y_E, z_E$) with which the initial first frame (B-frame) coincides, as shown in Fig. 3. The attitude of the quadrotor robot is described via the Euler angles yaw, pitch, and roll (ψ, θ, ϕ) .

This study examines the kinematics and dynamics of a quadrotor system, as illustrated in Fig. 3, which highlights the system's inertia, body coordinates, and the forces acting on it.

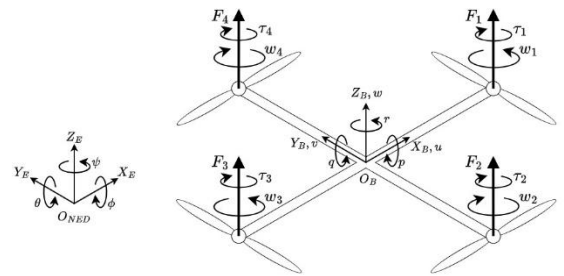


Fig. 3. The quadrotor Robot Body-frame (B-frame) and Earth-frame (E-frame) coordinates

To describe the quadrotor system, it is used several variables:

- Position Axes: x_k, y_k , and z_k ($k = E, B$) represent the position in the Earth frame (E-frame) and the Body frame (B-frame), respectively.
- Linear Velocity: u , v , and w indicate the linear velocities in the x_B, y_B , and z_B axes, of the B-frame.
- Angular Rates: p, q , and r axes denote the angular rates around the x_B, y_B , and z_B axes.
- Euler Angles: ϕ, θ , and ψ represent the roll, pitch, and yaw angles in the E-frame, respectively.

To simplify the modeling of a quadrotor, several key assumptions are typically made:

- The quadrotor's structure is assumed to be a rigid body to facilitate transformation calculations.
- The system is considered symmetric to simplify the calculation of the diagonal inertia matrix.
- The center of mass is assumed to coincide with the body-fixed frame, making it easier to identify the center of mass.
- Propellers are treated as identical and rigid, thereby avoiding the effects of blade flapping.
- Motors are assumed to be nearly identical, allowing for a generalized electric model.
- Thrust and drag are considered under perfect gas conditions with laminar airflow.
- Yaw angle will be under the condition ($-\pi < \psi < \pi$). Pitch angle and roll angle will be under the condition ($-\frac{\pi}{2} < \theta, \phi < \frac{\pi}{2}$) to avoid any non-singularity case for transformation matrix and unstable condition for the system.

III. MATHEMATICAL MODELLING

This section presents a detailed and comprehensive kinematic model for the quadrotor system, offering insights into its motion and orientation. Additionally, it provides a systematic dynamic model for the quadrotor UAV, capturing the complexities of its behavior under various forces and conditions.

A. Kinematics Model

To determine the orientation of a vehicle, methods such as trigonometric functions, Euler angles, and quaternions can be used. For simplicity, this paper will use Euler angles despite their singularity issues. Euler angles help us find the vehicle's final orientation relative to the body frame using an inertial-to-body frame transformation matrix. Matrix multiplication is convenient for transforming vectors.

The position of the vehicle can be described using x_E, y_E , and z_E coordinates, which represent the respective distances from a fixed origin in the inertial frame of reference. This frame of reference is typically aligned with the cardinal directions: North, East, and Down (O_{NED}). To represent the vehicle's orientation or attitude, the Euler angles ϕ, θ , and ψ are used. These angles are defined with respect to the body frame reference, which is a coordinate system centered at the vehicle's center of gravity (O_B). Fig. 4 illustrates these frames of reference for enhanced visualization.

When rotating around one body frame axis, the other two axes are displaced relative to the inertial frame axes, while

the rotation axis remains parallel to its corresponding inertial axis. These rotated body frame axes can be represented as trigonometric function equations and subsequently expressed in matrix form.

- Around z-axis:

$$\begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = \begin{bmatrix} c_\psi & s_\psi & 0 \\ -s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} = R(\psi) \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} \quad (4)$$

- Around y-axis:

$$\begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = \begin{bmatrix} c_\theta & 0 & -s_\theta \\ 0 & 1 & 0 \\ s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} = R(\theta) \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} \quad (5)$$

- Around x-axis:

$$\begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & s_\phi \\ 0 & -s_\phi & c_\phi \end{bmatrix} \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} = R(\phi) \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} \quad (6)$$

By combining these three matrices in order (as shown in Fig. 4), the inertial-to-body frame transformation matrix is obtained. The order of operations is crucial.

$$\begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = R(\psi).R(\theta).R(\phi) \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} = R' \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} \quad (7)$$

$$R' = \begin{bmatrix} c_\theta c_\psi & c_\theta s_\psi & -s_\theta \\ c_\psi s_\theta s_\phi - c_\phi s_\psi & s_\psi s_\theta s_\phi + c_\phi c_\psi & c_\theta s_\phi \\ c_\psi s_\theta c_\phi + s_\phi s_\psi & s_\psi s_\theta c_\phi - s_\phi c_\psi & c_\theta c_\phi \end{bmatrix}$$

Multiplying these matrices results in a single transformation matrix R' for the inertial-to-body frame transformation.

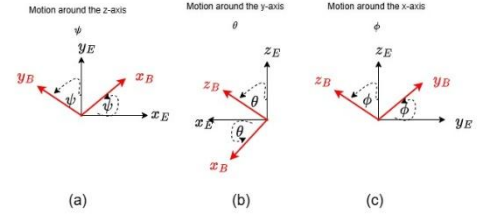


Fig. 4 Euler Angle Transformation

Using coordinate transformation, angular velocity can be represented as the derivative of the Euler angles. This relationship helps in understanding the rotational dynamics of the vehicle. The angular velocity in terms of the Euler angles (ϕ, θ , and ψ) and their time derivatives ($\dot{\phi}, \dot{\theta}$, and $\dot{\psi}$) can be derived.

The angular velocity components in the body frame are related to the time derivatives of the Euler angles as follows:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = R(\phi).R(\psi) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + R(\phi). \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

$$= T' \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$T' = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & c_\theta s_\phi \\ 0 & s_\phi & c_\theta c_\phi \end{bmatrix}$$

The actual translational and rotational motions occur in the B-frame but can be transformed into the E-frame using the following kinematic equations:

$$v_E = R v_B \quad (9.a)$$

$$w_E = T w_B \quad (9.b)$$

Here, $v_E = \dot{\xi}$, $w_E = \dot{\eta}$, $v_B = [u \ v \ \omega]^T$ represents the linear velocity vector in the B-frame, and $w_B = [p \ q \ r]^T$ represents the rotational velocity vector. The matrices $R = R'^{-1} \in SO(3)$ and $T = T'^{-1} \in SO(3)$ are the rotation and transformation matrices that convert B-frame motions to the E-frame, defined as follows:

$$R = \begin{bmatrix} c_\psi c_\theta & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix} \quad (10)$$

$$T = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \\ 0 & c_\theta & s_\theta \end{bmatrix}$$

B. Dynamic Model

The quadrotor's dynamics can be effectively modeled using the Newton-Euler method, which combines Newton's second law for translational motion and Euler's equations for rotational motion. This comprehensive approach provides a detailed understanding of the forces and torques acting on the quadrotor, enabling precise control and navigation. By integrating translational and rotational equations of motion, it could be to simulate and control the quadrotor's behavior with high accuracy, addressing both the forces and torques that influence its flight. This mathematical model is fundamental for developing advanced control algorithms and enhancing the autonomous capabilities of quadrotor UAVs.

The quadrotor's dynamics are described by two nonlinear models: translational dynamics (governing linear motion) and rotational dynamics (governing orientation). Additionally, it worth to mention also rotor dynamics is important to control the speed of quadrotor. Based on Newton 2nd law $F_{net} = ma$, let's implement that to quadrotor translation motion as follows:

$$m\ddot{r} = F_{th} - F_g - F_{cf} - F_{dg} + F_{ext} \quad (11)$$

where m is the mass of quadrotor, $r = [x_B, y_B, z_B]^T$, and F_{th} is the actual force generated by quadrotor's rotors called thrust force. This force affects on z-axis defined as follows

$$F_{th} = \begin{bmatrix} 0 \\ 0 \\ \sum_{i=1}^4 b_i \omega_i^2 \end{bmatrix}, \text{ where } b_i \text{ is lift constant } \omega_i \text{ is rotor}$$

angular speed as shown in Fig. 3. The F_g represents gravity force which acting opposite of lifting in z-axis which defines

$$\text{as follows } F_g = R' m g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \text{ and } F_{cf} \text{ is centrifugal force acting}$$

opposite the motion defined as follows $F_{cf} = v_B \times (m v_B)$. While F_{dg} represents the drag force which is Resistance caused by air acting opposite to the motion defined as follows

$$F_{dg} = \begin{bmatrix} k_x \dot{x} \\ k_y \dot{y} \\ k_z \dot{z} \end{bmatrix}, \text{ where } k_i, i = \{x, y, z\} \text{ is drag coefficient. The}$$

F_{ext} is external forces such as wind or collision impacts, also known as external disturbances.

For simplicity most researchers and engineering ignore some forces to let model be much simpler and applicable to hardware such as the centrifugal forces is totally removed from model because the effect of this force in inertial frame (Earth-fixed frame) is neglectable. Also, others ignore drag and external forces, too. Here, the proposed model for translation dynamics is compound all those forces in single function called translation unknown dynamics $f_{tr}(\cdot) = F_{ext} - F_{cf} - F_{dg}$, rather than neglect them. In this way, the designer has a more comprehensive model for the system. So, if the any part of this function is represented in the actual system that causes bias or error. Thus, the designers who eliminate this part need a very strict model and high constraint operation condition. Based on Eq. (9), the translational dynamics of quadrotor in E-frame as follows:

$$\ddot{\xi} = \frac{R}{m} F_{th} - G + \overline{f_{tr}}(\cdot) \quad (12.a)$$

where $G = \frac{R}{m} F_g = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$, and here $\overline{f_{tr}}(\cdot) = \frac{R}{m} (F_{ext} - F_{cf} -$

$F_{dg})$.

Thus, it possible to rewrite the translational dynamics as follows:

$$\ddot{\xi} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} \frac{F_t}{m} (c_\phi s_\theta c_\psi + s_\phi s_\psi) \\ \frac{F_t}{m} (c_\phi s_\theta s_\psi - s_\phi c_\psi) \\ \frac{F_t}{m} (c_\phi c_\theta) \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} + \begin{bmatrix} f_{tr_x} \\ f_{tr_y} \\ f_{tr_z} \end{bmatrix} \quad (12.b)$$

where $F_t = \sum_{i=1}^4 b_i \omega_i^2$ is total thrust as discussed previously. The 2nd newton law for rotational motion defined as $I \dot{\omega} = M_{net}$, where $I \omega$ is inertia multiplied by angular acceleration, and M_{net} is net moment. Same methodology implement for rotational dynamics as follows:

$$I \dot{v}_B = M_\tau - M_{cpt} - M_{gyr} + M_d \quad (13)$$

where $I = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$ is inertia matrix of quadrotor

where I_{xx} is moment of inertia around x_E , I_{yy} is moment of inertia around y_E , and I_{zz} is moment of inertia around z_E . While M_τ is actual torques moment produces by quadrotor's rotors which defined as follows $M_\tau = [l\tau_\phi, l\tau_\theta, k_c \tau_\psi]^T = [lb(\omega_4^2 - \omega_2^2), lb(\omega_4^2 - \omega_3^2), k_c(\omega_4^2 - \omega_3^2 + \omega_2^2 - \omega_1^2)]^T$. Where l is the length between rotor and center of quadrotor (center of mass), b is thrust coefficient, and k_c is torque coefficient also it called force to moment scaling factor. While M_{cpt} is centripetal moment defined as $M_{cpt} = v_B \times I v_B = [(I_{zz} - I_{yy})qr, (I_{zz} - I_{xx})pr, (I_{yy} - I_{xx})pq]^T$, and M_{gyr} is gyroscopic moment or effect defined as $M_{gyr} = (I_r v_B) \times ([0, 0, 1]^T w_r)$. Where I_r is rotor moment of inertia, and w_r is directional summation of rotational speed for each individual motor. While M_d is disturbances moment that comes from drag effects, blade flapping effects, ceiling and ground effects, or any external disturbances that effects the rotors torques or quadrotor.

To simplify the rotational dynamics, many researchers and engineering ignoring the gyroscopic effects due to I_r is very small value, and others also neglect external disturbances and

centripetal forces. Here, the proposed model for rotational dynamics composed all those forces and moment in single function called rotational unknown dynamics defined as follows: $f_{ro}(\cdot) = M_d - M_{g_{yr}}$, rather than neglect them. In this way, the designer has a more comprehensive model for the system. So, if the any part of this function is represented in the actual system that cause bias or error. Thus, the designers who eliminate this part need very strict model and high constraint operation condition. The Eq. (13) could be rewritten as follows:

$$\dot{v}_B = I^{-1}(M_\tau - M_{cpt} + f_{ro}(\cdot)) \quad (14.a)$$

Based on Eq. (9), the rotational dynamics of quadrotor in E-frame as follows:

$$\ddot{\eta} = \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} = \begin{bmatrix} \frac{l}{I_{xx}} & 0 & 0 \\ 0 & \frac{l}{I_{yy}} & 0 \\ 0 & 0 & \frac{k_c}{I_{zz}} \end{bmatrix} \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} - \begin{bmatrix} \frac{lk_\phi}{I_{xx}} & 0 & 0 \\ 0 & \frac{lk_\theta}{I_{yy}} & 0 \\ 0 & 0 & \frac{k_c k_\psi}{I_{zz}} \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \overline{f_{ro}}(\cdot) \quad (14.b)$$

here $\overline{f_{ro}}(\cdot) = [f_{ro_x}, f_{ro_y}, f_{ro_z}]^T$, which is unknown rotational dynamics around each axis $\{x, y, z\}$. While k_ϕ , k_θ , and k_ψ are the air drag coefficient effects on each angle.

IV. CONTROL STRATEGY

The dynamic model reveals that the quadrotor's attitude is primarily influenced by the input moments, while its translational dynamics are governed by the angles and total thrust forces, as described in Equations (12) and (14) and illustrated in Fig. 3 and 5. Notably, the translational subsystem depends on the angles, yet these angles and their derivatives are independent of the translational components, as shown in Equation (14). This emphasizes the importance of the attitude controller for ensuring system control and stability. Additionally, since the core actuator in a quadrotor is the BLDC motor, understanding rotor dynamics is essential for effectively controlling the quadrotor's speed.

The quadrotor control problem is generally divided into three components:

- Position Control: Regulates the quadrotor's position.
- Attitude Control: Manages the quadrotor's orientation and stability.
- Speed Control: Regulates the quadrotor's rotor speed.

Typically, the speed controller cascades with the attitude controller, forming what is known as the inner loop, while the position controller forms the outer loop, as shown in Fig. 5. From previous analysis, it is possible to drive relationship between the command to quadrotor BLDC motors and rotor speed as follows.

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} b & b & b & b \\ 0 & -lb & 0 & lb \\ -lb & 0 & lb & 0 \\ k_c & k_c & k_c & k_c \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} = \mathcal{M} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \quad (15)$$

The control inputs of the quadrotor are generated by the thrust forces produced by the four rotors. These thrusts collectively create both the total lift force and the control moments required to maneuver the vehicle. The relationship between the rotor thrusts and the resulting forces and moments can be represented by the force-moment matrix $\mathcal{M}$. This matrix maps the individual rotor inputs to the total thrust and rotational moments acting on the quadrotor. The specific structure of $\mathcal{M}$ depends on the adopted modeling assumptions and the physical configuration of the quadrotor, including the rotor arrangement, arm length, thrust coefficients, and drag coefficients.

Here, an additional assumption is added that all rotors have same thrust coefficient and that is logic. Where it should be using same type of BLDC motor. Thus, the desired speed for BLDC motors that need to be controlled as follows:

$$\Omega_d = (\mathcal{M}^{-1}U)^{\frac{1}{2}} \quad (16)$$

Where $\Omega_d = [\omega_{d1}, \omega_{d2}, \omega_{d3}, \omega_{d4}]^T$, and $U = [u_1, u_2, u_3, u_4]$. In this paper, it is assumed that a speed controller is available, so the focus is on the dynamics and control of the quadrotor system. As previously mentioned, there are various techniques proposed for controlling the quadrotor drone. Based on Equations (12) and (14), suggested methods for controlling the position and attitude of the quadrotor include estimating and rejecting unknown dynamics, such as through Active Disturbance Rejection Control (ADRC) and Model-free Control (MFC) or enhancing the control system's tolerance using techniques like Disturbance Observer-based Control (DOBC) and Model-reference Adaptive Control (MRAC). Additionally, a PID controller can be employed under strong assumptions and constraints, effective only in specific conditions or flight modes.

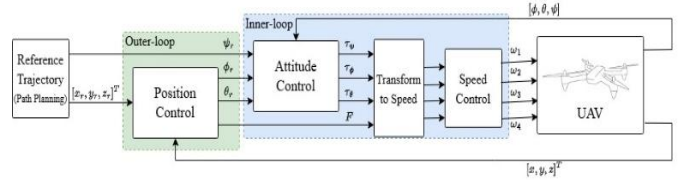


Fig. 5 The overall structure of the control system for the quadrotor

V. CONCLUSION

This paper provided a comprehensive and systematic mathematical model for quadrotor UAVs, covering both the kinematics and dynamics using the Newton-Euler method, which is technically sound and accessible for understanding quadrotor behavior across various flight modes, including vertical take-off, horizontal flight, and aggressive maneuvers. This model serves as a valuable tool for researchers, offering clear guidance on deriving mathematical formulations crucial for quadrotor analysis while addressing complexities such as nonlinearity and coupling effects. It also suggested suitable control strategies, such as Active Disturbance Rejection Control (ADRC), Model-Free Control (MFC), and robust adaptive techniques, to enhance system performance while managing external disturbances and uncertainties. Future work should focus on controller design, simulation, and practical implementation to extend the operational capabilities of quadrotors, particularly in aggressive

maneuvering and challenging environments, thereby expanding their versatility and utility in diverse applications.

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